TECHNICAL NOTE

Governance costing model: technical note (v018.1)

Emanuel Marques Queiroga · 10 November 2026 · AI in Education Procurement Tool · v018.1

Summary

The governance cost is estimated option by option, with explicit calculations and parameters that users can see and adjust in the tool. Each option selected in the plan has its own calculation, and the total cost is the sum of the options plus three additional costs calculated on top of them. Every amount is split into investment (CapEx), operations (OpEx) and usage, which is metered cloud consumption.

Most of the cost comes from people's work. For 31 options, the tool uses a reference team, with the roles needed, the number of people, each person's time share and the months of work, and users can add roles. The monthly cost of each role starts from the country's average earnings for the corresponding occupational group (ILOSTAT, ILO), updated to 2026 with the country's inflation (IMF) and increased by the mandatory employer contribution (SSA and ISSA). Roles that require AI, data or digital security skills receive a 28% wage premium, measured across 1.3 billion job postings (Lightcast, 2025).

In the data, infrastructure and technological independence options (challenge 2), the purchased parts are added to the team: cloud consumption at public list prices, costs from the official X-Road calculators, a sovereign cloud model per 100,000 inhabitants with build and contract paths, and entered amounts. Travel and in-person activities (25% of people's work), overhead (15% of people's work and travel) and an administrative fee (7% of the options and travel) are applied to the cost of the options, each with a public source and a question that indicates whether it applies to the plan, plus a question on training people, whose number is set by whoever prepares the plan. Training is counted by components: course creation, instructors, room and meals on in-person days, instructors' travel and coordination. When full-time staff, training and the size of the challenge 2 systems have no entered quantity, the tool uses reference estimates. The tool shows the economic cost of the plan and, separately, the new outlay, which excludes the work of existing civil servants.

Scope and cost structure

Governance has 3 challenges, 9 categories and 37 options, and each option has a calculation defined by its cost type. Category 3.2 has no options, so the category itself is the cost line.

Cost type	Where it applies	Calculation	Source of the unit value
Reference team	31 options in categories 1.1, 1.2, 1.3, 2.1, 2.2, 2.3 and 3.3	Sum over roles: people × time share × months × monthly cost	Country earnings (ILOSTAT) × inflation (IMF) × contributions (SSA and ISSA) × specialist premium, where applicable
Full-time staff	4 options in category 3.1	People (default: 2 per profile) × monthly cost × plan duration	Country earnings, group 2, with the same treatment; premium in the AI, data and IT options
Training	Category 3.2	Course creation + instructors + room and meals + instructors' travel + coordination (default: people in the plan, 20 hours, half in person)	ATD (2017); UKCAMS (2026); World Bank (price level)
Purchased parts and consumption	Options in categories 2.1, 2.2 and 2.3	Consumption × public price; X-Road working days × daily rate; sovereign cloud model; entered amount	Azure and Google Cloud; NIIS; GTMI, IMF, World Bank, KPMG, BCG and ODC-Noord; user
Percentage	Travel, overhead and administrative fee	Percentage × base	Global Fund, 2 CFR 200.414, UNSDG policy
Training of people	Additional costs, if whoever prepares the plan wants it	The same calculation as 3.2, with the people entered	ATD (2017); UKCAMS (2026); World Bank (price level)

General equation and notation

The total cost is the direct cost plus the three additional costs. The direct cost includes only the options that already have an amount; the others are shown as "missing amount" and enter the total once they are priced. People's work P is the direct cost without the purchased parts of challenge 2 and without training; Q is the training of people included in the additional costs; and the total cost is split into investment K, operations X and usage U.

$$C = D + V + O + T + Q, \qquad D = \sum_{o} C_o, \qquad C = K + X + U$$

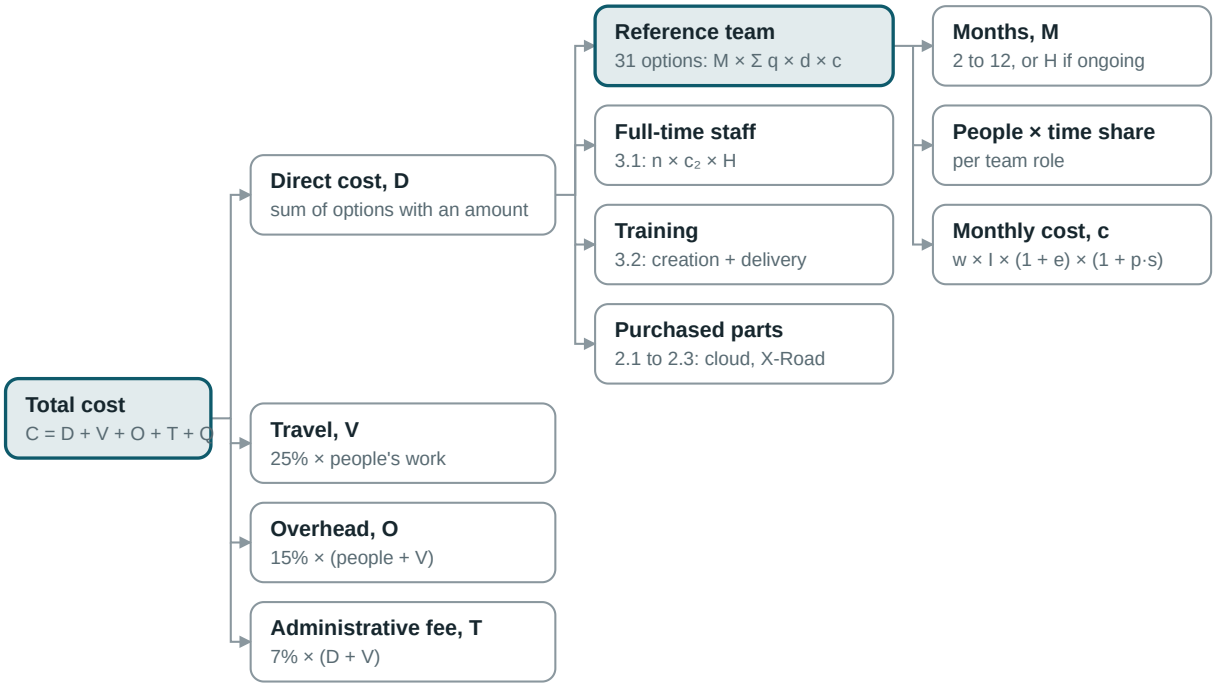


Figure 1. Structure of the governance cost, from the total to the parameters.

Symbol	Meaning	Unit
C	Total cost of the governance plan	plan currency
D	Direct cost: sum of the options that already have an amount	plan currency
P	People's work: direct cost without the purchased parts (infrastructure, equipment, licences and cloud services)	plan currency
K, X, U	Investment (CapEx), operations (OpEx) and usage (metered cloud consumption) of the plan	plan currency
Q	Training of people in the additional costs	plan currency
C_o	Cost of option o	plan currency
V, O, T	Travel and in-person activities, overhead and administrative fee	plan currency
H	Plan duration (default 12)	months
M	Working months of the option (one-off) or H (ongoing)	months
q, d	People and time share of a team role	people; share of time
w	Average monthly earnings of the occupational group in the country, in the data year	plan currency per month
I	Inflation update factor, from the data year to 2026	index
e	Employer contribution on payroll	share
p, s	Specialist premium and specialist-role indicator (1 or 0)	share; 0 or 1

Symbol	Meaning	Unit
x	Exchange rate: plan currency per US$	currency per US$
c_2*	Monthly cost of a group 2 professional with the specialist premium	plan currency per month
ρ	Cloud price per GB per month, per GB transferred or per machine hour	US$
N, m, λ	Sovereign cloud: population ÷ 100,000, digital maturity adjustment and education share	units; index; share

Options with a reference team

The cost of an option with a reference team is the team's monthly cost multiplied by the months of work. The monthly cost adds up, role by role, the number of people, the time share and the monthly cost of one person in that role.

$$C_o = M \sum_r q_r \, d_r \, c_r, \qquad c_r = w_{g(r)} \cdot I \cdot (1 + e) \cdot (1 + p \, s_r)$$

Months of work

One-off options, such as a regulation, a guideline, a sandbox or a data inventory, have their own reference months, from 2 to 12. Ongoing options, such as supervision, committees, interministerial coordination, operation of the data exchange platform and open-source maintenance, last the whole plan, so M = H.

Country earnings

Earnings w are the monthly average of employees in the role's ISCO-08 major group (1, managers; 2, professionals; 3, technicians and associate professionals; 4, clerical support workers), from the ILOSTAT indicator, for both sexes and the latest year available since 2015. When the plan uses the country's currency, the tool uses the local-currency value; otherwise, it converts the US$ value published by the ILO with the exchange rate x.

For countries without their own data, w is the US$ median of countries with data in the same World Bank region and income group, with at least three countries, or the median of the income group. In the absence of any estimate, the world median of countries with their own data is used. Values below US$30 a month are discarded as recording errors.

Inflation update

Because earnings data refer to different years, each value is updated from the data year to 2026 by the product of annual consumer price inflation in the following years, according to the IMF World Economic Outlook (with an estimate for 2025 and a projection for 2026). Earnings in local currency are updated with the country's inflation; earnings converted from US$, with United States inflation.

$$I = \prod_{t=a+1}^{2026} (1 + \pi_t)$$

Employer contributions

The factor 1 + e includes the mandatory employer social security contribution, as a percentage of payroll, from table 4 of Social Security Programs Throughout the World (SSA and ISSA; Africa and the Americas, 2019; Europe and Asia and the Pacific, 2018). Additional benefits set by contract or by specific legislation can be included by the user in the monthly cost of each role.

Specialist premium

Professionals with AI, data and digital security skills earn above the average of their occupational group. The model applies to these roles (s = 1, marked with an asterisk in the team table) a premium p of 28%, the salary difference between postings that do and do not require AI skills across 1.3 billion job ads (Lightcast, 2025). This is a conservative value: PwC (2025) estimated an average premium of 56% for jobs requiring AI skills, compared with 25% the year before.

Editable values and added roles

In the tool, users can change the months of a one-off option, the number of people in each role and the monthly cost of each role. Users can also add roles to any team, with a name, occupational profile, people, time share, specialist premium and monthly cost; an added role enters the team sum through the same equation. An entered value replaces the estimate,

and the line moves from "estimated" to "entered" once all its values have been entered.

Full-time staff and training

The options without a reference team in categories 3.1 and 3.2 use simpler calculations. When the user does not enter the quantity, the tool uses reference estimates (grade C): two people per profile for full-time staff, the minimum pair to keep the work going during leave and absences, and, for training, the people in the reference teams and staff chosen in the plan (with no teams in the plan, a class of 25 people), with 20 hours each, half in person. In the four full-time staff options (3.1), the cost is the number of people n times the monthly cost of group 2, calculated as above, times the plan duration; the AI, data and IT options receive the specialist premium.

$$C_o = n \cdot w_2 \cdot I \cdot (1+e) \cdot (1+p\,s_o) \cdot H$$

For staff training (3.2), the cost adds up what a course actually costs. Training blends in-person sessions, taken to the participants' region, with live online sessions, with an in-person share θ (default 50%). Course creation, which is investment, takes 38 hours of development per in-person hour and 28 per live online hour, according to the ATD survey (2017), at the specialist's hourly cost c_h = 12 c_2* ÷ 2,080. Delivery, which is operations, has two instructors per class of 25, paid for the teaching and for the same time in preparation; room, coffee breaks and lunch on 7-hour in-person days, for participants and instructors, at the day delegate rate u_e; accommodation and travel for the instructors, at the residential rate u_r and the travel cost u_t; and 15% for coordination, materials and certification (κ). The daily rates are the United Kingdom averages for 2025, £51 and £187 including VAT (UKCAMS, 2026), adjusted by the country price level ℓ (private consumption PPP conversion factor divided by the exchange rate, World Bank) and converted at the pound exchange rate; travel is US$100 × ℓ (grade C).

$$\tau = \lceil n_f/25 \rceil, \quad a = \lceil \theta h_f/7 \rceil, \quad C_o = h_f\,[38\,\theta + 28\,(1-\theta)]\,c_h + (1+\kappa)\,[4\,\tau h_f c_h + (n_f + 2\tau)\,a\,u_e + 2\tau\,(a\,u_r + u_t)]$$

Participants' time, n_f × h_f × 12 c_2 ÷ 2,080, is shown separately and is not part of the total, because it is not new spending. Training is outside the base for travel and overhead, because it already has its own logistics and coordination. With one class, course creation is most of the cost; with many, delivery dominates. As a market reference, IAPP's official AI governance training costs US$1,195 for 13 online hours, about US$92 per hour per person.

Data, infrastructure and technological independence (challenge 2)

The 13 options of challenge 2 add, when there is people's work, the reference team and the purchased parts j. Cloud consumption is counted by volume, using the public price list of a major provider, and what depends on the supplier, such as network links, commercial licences, equipment and support contracts, is entered as an amount. Without an entered quantity, reference estimates apply (grade C): 5 systems to integrate; for data stored and backed up, 10 MB per student enrolled in primary and secondary education (World Bank, SE.PRM.ENRL and SE.SEC.ENRL; without data, 10 TB); 4 virtual machines per million students, at least 4; and, for the sovereign cloud, the contract path. Only the country population, when missing, leaves the option pending.

$$C_o = C_o^{\mathrm{eq}} + \sum_j B_{o,j}$$

Option	Part	Calculation	Class
2.1 Data inventory and curation	Reference team (6 months)	Sum over roles	OpEx
	Ongoing curation	curators (default: 2) × c_2* × H	OpEx
	Data catalogue (CKAN)	1 virtual machine × ρ_vm × 730 × H × x	Usage
2.1 Access controls and retention rules	Reference team (6 months)	Sum over roles	OpEx
	Access control (Keycloak)	2 virtual machines × ρ_vm × 730 × H × x	Usage
	Records archive (optional)	TB × 1,024 × ρ_arq × H × x	Usage
	Commercial licence (optional)	entered annual amount × H ÷ 12	OpEx
2.1 Data interoperability and portability	Reference team (6 months)	Sum over roles	OpEx

Option	Part	Calculation	Class
	Integration of each system	n_s × 44 days × 12 c_2* ÷ 260	CapEx
	Servers and certificates	n_s × € 6,480 × H ÷ 12	OpEx
	Maintenance of the integrations	n_s × 0.1 × c_2* × H	OpEx
2.1 Shared public data infrastructure	Operations team (1.5 people, ongoing)	Sum over roles	OpEx
	Platform deployment	655 days × 12 c_2* ÷ 260	CapEx
	Servers, security modules and certificates	€ 62,160 × H ÷ 12	OpEx
2.2 Sovereign cloud, build	Investment	λ K_nac x	CapEx
	Facilities and platform team	λ [r K_nac x + 12 c_2* (n_0 + n_1 N m)] H ÷ 12	OpEx
2.2 Sovereign cloud, contract	Deployment and migration team (6 months)	Sum over roles	CapEx
	Contracted service	λ [K_nac x ÷ L + r K_nac x + 12 c_2* (n_0 + n_1 N m)] (1 + φ) H ÷ 12	OpEx
2.2 Secure hosting and data storage	Storage	TB × 1,024 × ρ_arm × H × x	Usage
	Data transfer out of the cloud (optional)	TB per month × 1,024 × ρ_saida × H × x	Usage
2.2 Compute and network infrastructure	Virtual machines	n_vm × ρ_vm × 730 × H × x	Usage
	Network links (optional)	entered annual amount × H ÷ 12	OpEx
	Own network equipment (optional)	entered total amount	CapEx
2.2 Backup and service continuity	Reference team (2 months)	Sum over roles	OpEx
	Backup copies (3-2-1 rule)	TB × 1,024 × (ρ_frio + ρ_arq) × H × x	Usage
	Standby environment (optional)	entered annual amount × H ÷ 12	OpEx
2.3 Open-source adoption and maintenance	Reference team (ongoing)	Sum over roles	OpEx
	Support contract (optional)	entered annual amount × H ÷ 12	OpEx
2.3 Technology transfer	Reference team (8 months)	Sum over roles	OpEx
	Licensing or fee (optional)	entered total amount	CapEx
2.3 Local customisation	Reference team (8 months)	Sum over roles	CapEx
2.3 Migration and exit	Reference team (6 months)	Sum over roles	CapEx
	Data retrieval (optional)	TB × 1,024 × ρ_saida × x	CapEx
	Parallel operation (optional)	months × monthly cost of the current system	OpEx

Cloud prices

Prices come from the Microsoft Azure public price list (Retail Prices API, retrieved 10 November 2026), in dollars, converted with the exchange rate x: hot-tier storage, US$0.0208 per GB per month (Google Cloud Storage charges about US$0.020); infrequently accessed copy, US$0.0152; archive copy, US$0.00099; data transfer out of the cloud, US$0.087 per GB; and a virtual machine with 2 cores and 8 GB, US$0.115 per hour, the median of the provider's 68 regions, with 730 hours per month. These metered consumptions form the usage class, separate from operations. Backup copies follow

the 3-2-1 rule recommended by CISA: besides the main data, an infrequently accessed copy and an archive copy, in another location. In the European Union, the Data Act bans switching charges from 12 January 2027, which may bring data retrieval in a migration to zero.

X-Road

Interoperability and the public data infrastructure follow the official X-Road cost calculators (NIIS, 2021), the secure data exchange platform used by Estonia and other countries. Deploying the central platform takes 655 working days and connecting each system 44 days, at a specialist's daily rate (12 c_2* ÷ 260 days); servers, security modules and certificates cost €62,160 per year for the central platform and €6,480 per year per connected system, converted at the euro rate; operating the central platform takes 1.5 people and maintaining the integrations 0.1 person per system.

Sovereign cloud

The sovereign cloud, under the country's jurisdiction and control, has a simplified model per 100,000 inhabitants, with two paths. The unit is the whole country, because the cloud serves the whole government, and education accounts for the share λ, which is education spending as a percentage of government spending (World Bank, SE.XPD.TOTL.GB.ZS) and can be changed in the plan. The country is classified into the three IMF groups (advanced economy, emerging market and middle-income economy, and low-income developing country), and the government's digital maturity enters through the Core Government Systems Index (CGSI) component of the World Bank's GovTech Maturity Index 2025; a country without a CGSI receives the median of its group.

If the government builds the infrastructure, the national investment has a floor F, which exists even in small countries, and a part proportional to the population, adjusted by the IMF group and by digital maturity, because a more mature government moves more systems to the cloud. Annual operations add facility maintenance, 2.6% of the investment per year (KPMG, 2026), and the platform team, with a 24-hour floor of 20 people and 1.1 people per 100,000 adjusted inhabitants, an order of magnitude consistent with the roughly 200 professionals of the Dutch government cloud (ODC-Noord, 2026).

$$K_{\text{nac}} = F + k_g\, N\, m, \qquad m = \tfrac{1}{2}\,(1 + \text{CGSI}), \qquad X_{\text{nac}} = r\, K_{\text{nac}} + 12\, c_2^{*}\,(n_0 + n_1 N m)$$

If the government contracts a provider, it pays each year the cost equivalent to building, with the investment spread over a 10-year useful life, plus the price premium of sovereign cloud over regular cloud, 20% (BCG, 2025, estimates 15% to 30%). The investment in this path is the reference team for deployment and migration to the provider.

$$B_{\text{contract}} = \lambda \left[\frac{K_{\text{nac}} x}{L} + r\, K_{\text{nac}} x + 12\, c_2^{*}(n_0 + n_1 N m) \right] (1 + \varphi)\, \frac{H}{12}$$

The coefficients of the build path were calibrated with sourced public investments in government cloud or data centres. The US$9 million floor comes from the two smallest cases (Estonia and Moldova), and the part per 100,000 inhabitants from the average of the cases in each group after deducting the floor: US$2.19 million in advanced economies and US$300,000 in the emerging and low-income groups, which use the same value because the cases do not show lower technology costs in lower-income countries. The table compares the observed value with what the model computes for the same country and population. The Sri Lanka case was left out of the calibration, because phase 1 of the government cloud used existing data centres.

Country	IMF group	Population (millions)	Investment	Observed (US$ millions)	Model (US$ millions)	Model ÷ observed
Estonia	advanced economy	1.37	€11.09M	12.20	38.61	3.16
Switzerland	advanced economy	8.87	CHF 319.40M	376.60	181.49	0.48
Indonesia	emerging market and middle-income economy	281.19	€164.70M	173.00	778.84	4.50
Sri Lanka	emerging market and middle-income economy	22.97	LKR 751.80M	4.20	72.05	outside the calibration
Moldova	low-income developing country	3.07	$6.50M	6.50	17.54	2.70
Senegal	low-income developing country	18.08	F CFA 46B	83.00	44.54	0.54

Country	IMF group	Population (millions)	Investment	Observed (US$ millions)	Model (US$ millions)	Model ÷ observed
Kenya	low-income developing country	55.34	CN¥1.23B	177.00	161.10	0.91
Bangladesh	low-income developing country	171.47	BDT 15.17B	193.00	506.13	2.62
Zambia	low-income developing country	20.72	$75.00M	75.00	62.34	0.83

In the eight cases used, the model falls between half and four and a half times the observed value, and tends to overestimate projects that are only one phase of a larger programme, such as the first of Indonesia's four national data centres. The model is preliminary (grade C), and the coefficients should be recalibrated when more sourced cases are available.

Additional costs

Travel applies to people's work P, and overhead to that work plus travel; the administrative fee applies to the direct cost plus travel, and neither overhead nor the fee enters the other's base. Purchases of infrastructure, equipment, licences and cloud services are excluded from the travel and overhead bases, as in the sources of both percentages, which exclude the purchase of products (Global Fund) and equipment and capital expenditures (2 CFR 200.1, modified total direct costs). In the options outside challenge 2, P is the whole cost, and the result is the same as in the previous version.

Training of people Q has its own question: if whoever prepares the plan wants to include the cost, they enter how many people will be trained, and the tool applies the same calculation as category 3.2, with the hours (default 20) and the in-person share (default 50%) editable. Course creation is investment and the rest is operations; Q does not enter the base of the other additional costs; if the plan also has category 3.2, the people counted here should be different ones.

$$Q = \delta_Q \, n_Q \, h_Q \, 12 \, x$$

$$V = p_V \, P, \qquad O = p_O \, (P + V), \qquad T = p_T \, (D + V)$$

Cost	Default	Base	Source of the default
Travel and in-person activities (fares, per diems, workshops, technical meetings)	25%	P	Global Fund: travel-related costs are 24.7% of programme budgets in cycle 7, excluding health products (27.6% in cycle 6; 28.2% in cycle 5)
Overhead (premises, IT, HR, accounting, legal)	15%	P + V	United States, 2 CFR 200.414(f): up to 15% of modified total direct costs without a negotiated rate (2024 revision)
Administrative fee (charged by whoever executes the funds)	7%	D + V	UNSDG policy for UN pooled funds: 7% of programme costs

Each additional cost has a question that indicates whether it applies to the plan. Without an answer, the default applies; the answer that excludes the cost sets its percentage to zero.

Question	Answer that excludes the cost	Effect
Will the plan have technical meetings, workshops or travel?	No	p_V = 0; overhead then applies to P only, and the fee to D only
Does the institution already cover indirect costs?	Yes, it already does	p_O = 0
Who will execute the funds?	The agency itself	p_T = 0; with a UN agency, 7%; with a support foundation, the contract fee (in Brazil, up to 15%, Decree 9,283/2018)

Investment, operations and usage

Each part of each option is investment (CapEx), operations (OpEx) or usage, under a simple accounting rule. Investment is the purchase of assets and the deployment of systems, platforms, integrations and migration, which create something that lasts beyond the plan: the sovereign cloud teams (in the contract path), local customisation and migration, the integration of systems and the deployment of X-Road, the investment of the built sovereign cloud, network equipment, technology transfer licensing, data retrieval and the creation of training courses. Usage is metered cloud consumption: storage, virtual machines (including those of the data catalogue and access control), backup copies, the records archive and data transfer. Everything else is operations: the work on regulations, guidelines, supervision and staff, the delivery of courses, licences and contracted services. Additional costs follow the proportion of their base: K_P in people's work, which has no usage, for travel and overhead, and K_D and U_D in the direct cost, for the fee. The plan's executive summary shows the three types by category and option, with the total.

$$K = \sum_{o} K_o + \frac{K_P}{P}\,(V + O) + \frac{K_D}{D}\,T, \qquad U = \sum_{o} U_o + \frac{U_D}{D}\,T, \qquad X = C - K - U$$

Plan cost and new outlay

The plan cost measures the value of all the work required, including that of people who are already civil servants of the agency. To support budget planning, users can mark, in each team, the roles held by existing civil servants, including added roles, and the tool also shows the new outlay.

$$\text{New outlay} = C - \sum_{r \,\in\, \text{existing}} M\, q_r\, d_r\, c_r$$

Duration, exchange rate and totals

Governance has a single period of duration H (default 12 months, editable), and all costs are added up within it. A one-off option counts all its reference months, even if they exceed H, and the implementation roadmap flags these cases in its calendar.

The exchange rate x is the one entered in the case or, failing that, the ExchangeRate-API rate of the day (USD base), retrieved when the tool is opened; if that source does not respond, the currency-api rate of the day is used. X-Road prices, in euros, are converted at the euro rate. The example in this note uses the rate of 3 October 2026, rounded to four decimals (R$5.2223 per US$) so that the calculation can be checked by hand. Amounts in different currencies are never added together. Each line has one of three statuses, and the total shows the first two separately.

Status	When	In the total
Entered	All the line's values were entered by the user	Yes
Estimated	At least one value comes from an estimate (country earnings, reference team, cloud price, default percentage)	Yes, identified as estimated
Missing amount	A quantity, price, months value or required choice is missing	No; shown as pending

Reference team parameters

The 31 teams have 2 to 8 people; one-off options last 2 to 12 months and ongoing ones the whole plan. The asterisk marks roles with the specialist premium, and the last column names the public case used to check the order of magnitude.

Cat.	Option	Months	People	Full-time equivalent	Team: role (profile) people × time share	Comparison
1.1	AI safety and risk classification	8	7	5.5	Coordination (G1) 1 × 0.5; Digital law and regulation specialist* (G2) 2 × 1; AI and risk assessment specialist* (G2) 1 × 1; Education policy specialist (G2) 1 × 0.5; Public consultation analyst (G3) 1 × 1; Administrative support (G4) 1 × 0.5	

Cat.	Option	Months	People	Full-time equivalent	Team: role (profile) people × time share	Comparison
1.1	Children's rights and protection	6	6	3.5	Coordination (G1) 1 × 0.5; Children's rights specialist (G2) 1 × 1; Child development and education specialist (G2) 1 × 0.5; AI specialist* (G2) 1 × 0.5; Public consultation analyst (G3) 1 × 0.5; Administrative support (G4) 1 × 0.5	
1.1	Personal data protection and privacy	8	7	5.5	Coordination (G1) 1 × 0.5; Data protection law specialist* (G2) 2 × 1; Education data specialist* (G2) 1 × 1; Data protection officer* (G2) 1 × 0.5; Public consultation analyst (G3) 1 × 1; Administrative support (G4) 1 × 0.5	Kenya data protection office
1.1	Algorithmic fairness, non-discrimination, transparency, explainability and accountability	9	8	6.5	Coordination (G1) 1 × 0.5; AI ethics and fairness specialist* (G2) 1 × 1; Data scientist* (G2) 1 × 1; AI explainability specialist* (G2) 1 × 1; Fundamental rights specialist (G2) 1 × 1; Inclusive education specialist (G2) 1 × 0.5; Technical analyst (G3) 1 × 1; Administrative support (G4) 1 × 0.5	
1.1	Cybersecurity and incident reporting	6	6	4.5	Coordination (G1) 1 × 0.5; Information security specialist* (G2) 2 × 1; Digital law specialist* (G2) 1 × 0.5; Incident response analyst* (G3) 1 × 1; Administrative support (G4) 1 × 0.5	
1.1	Intellectual property and copyright	5	5	2.5	Coordination (G1) 1 × 0.25; Intellectual property specialist (G2) 1 × 1; Educational resources specialist (G2) 1 × 0.5; Generative AI specialist* (G2) 1 × 0.25; Administrative support (G4) 1 × 0.5	
1.2	Controlled testing environments	12	8	6.5	Coordination (G1) 1 × 1; AI specialist* (G2) 1 × 1; Regulation specialist (G2) 1 × 1; Educational assessment specialist (G2) 1 × 0.5; IT infrastructure and security specialist* (G2) 1 × 0.5; Pilot school liaison (G3) 1 × 1; Technical analyst (G3) 1 × 1; Administrative support (G4) 1 × 0.5	Norway sandbox
1.2	Regulatory supervision	ongoing	7	7	Coordination (G1) 1 × 1; AI supervision specialist* (G2) 3 × 1; Administrative law specialist (G2) 1 × 1; Technical analyst (G3) 1 × 1; Administrative support (G4) 1 × 1	AI Act, COM(2021) 206
1.2	Risk assessments	6	4	2.25	AI risk assessment specialist* (G2) 1 × 1; Data scientist* (G2) 1 × 0.5; Data protection specialist* (G2) 1 × 0.25; Technical analyst (G3) 1 × 0.5	Norway sandbox
1.3	Ethical AI guidelines for education	5	5	3	Coordination (G1) 1 × 0.25; AI ethics specialist* (G2) 1 × 1; Education specialist (G2) 1 × 1; Technical writer (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
1.3	AI procurement and vendor evaluation guidelines	5	6	3.5	Coordination (G1) 1 × 0.25; Public procurement specialist (G2) 1 × 1; AI specialist* (G2) 1 × 1; Legal specialist (G2) 1 × 0.5; Technical analyst (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
1.3	Age-appropriate AI use guidelines	4	5	2.5	Coordination (G1) 1 × 0.25; Child development specialist (G2) 1 × 1; Education specialist (G2) 1 × 0.5; AI specialist* (G2) 1 × 0.5; Administrative support (G4) 1 × 0.25	
1.3	Teacher and student AI-use policies	4	6	2.75	Coordination (G1) 1 × 0.25; Education specialist (G2) 1 × 1; Assessment and academic integrity specialist (G2) 1 × 0.5; Legal specialist (G2) 1 × 0.25; Technical analyst (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
1.3	AI risk management standards	5	5	3	Coordination (G1) 1 × 0.25; Risk management specialist (G2) 1 × 1; AI specialist* (G2) 1 × 1; Technical analyst (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
1.3	AI transparency and documentation standards	4	5	2.5	Coordination (G1) 1 × 0.25; AI specialist* (G2) 1 × 1; Legal specialist (G2) 1 × 0.5; Technical analyst (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
1.3	Data governance and security standards	5	5	3	Coordination (G1) 1 × 0.25; Data governance specialist* (G2) 1 × 1; Information security specialist* (G2) 1 × 1; Technical analyst (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	

Cat.	Option	Months	People	Full-time equivalent	Team: role (profile) people × time share	Comparison
1.3	AI quality assurance and evaluation guidelines	5	5	3	Coordination (G1) 1 × 0.25; Educational assessment specialist (G2) 1 × 1; AI specialist* (G2) 1 × 1; Data analyst* (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
2.1	Data inventory and curation	6	5	3.75	Coordination (G1) 1 × 0.5; Data governance specialist* (G2) 1 × 1; Data analyst* (G3) 2 × 1; Administrative support (G4) 1 × 0.25	
2.1	Access controls and retention rules	6	5	3.25	Coordination (G1) 1 × 0.5; Data protection specialist* (G2) 1 × 1; Information security specialist* (G2) 1 × 1; Digital law specialist* (G2) 1 × 0.5; Administrative support (G4) 1 × 0.25	
2.1	Data interoperability and portability	6	5	3.25	Coordination (G1) 1 × 0.5; Data and interoperability architect* (G2) 1 × 1; Education data specialist* (G2) 1 × 1; Data analyst* (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
2.1	Shared public data infrastructure	ongoing	2	1.5	IT infrastructure and security specialist* (G2) 1 × 1; Support technician (G3) 1 × 0.5	
2.2	Sovereign cloud	6	6	4.25	Coordination (G1) 1 × 0.5; IT architect* (G2) 1 × 1; IT infrastructure and security specialist* (G2) 2 × 1; Contract manager (G2) 1 × 0.5; Administrative support (G4) 1 × 0.25	
2.2	Backup and service continuity	2	2	1	IT infrastructure and security specialist* (G2) 1 × 0.5; Support technician (G3) 1 × 0.5	
2.3	Open-source adoption and maintenance	ongoing	4	3.5	Technical lead* (G1) 1 × 0.5; Developer* (G2) 2 × 1; Support technician (G3) 1 × 1	
2.3	Technology transfer	8	6	3.75	Coordination (G1) 1 × 0.5; IT specialist* (G2) 2 × 1; Contract manager (G2) 1 × 0.5; Technical trainer (G3) 1 × 0.5; Administrative support (G4) 1 × 0.25	
2.3	Local customisation	8	6	5	Coordination (G1) 1 × 0.5; Developer* (G2) 2 × 1; Curriculum specialist (G2) 1 × 1; Language and translation specialist (G2) 1 × 0.5; Tester (G3) 1 × 1	
2.3	Vendor dependency reduction	4	4	2.25	Coordination (G1) 1 × 0.25; Public procurement specialist (G2) 1 × 1; IT architect* (G2) 1 × 0.5; Legal specialist (G2) 1 × 0.5	
2.3	Migration and exit	6	6	4.25	Coordination (G1) 1 × 0.5; IT architect* (G2) 1 × 1; Developer* (G2) 2 × 1; Contract manager (G2) 1 × 0.5; Administrative support (G4) 1 × 0.25	
3.3	Governance committees	ongoing	2	1	Executive secretary (G2) 1 × 0.5; Administrative support (G4) 1 × 0.5	
3.3	Interministerial coordination	ongoing	3	2.5	Coordination (G1) 1 × 1; Public policy analyst (G2) 1 × 1; Administrative support (G4) 1 × 0.5	
3.3	Institutional governance structures	ongoing	8	8	Coordination (G1) 1 × 1; AI specialist* (G2) 1 × 1; Data specialist* (G2) 1 × 1; Legal specialist (G2) 1 × 1; Education specialist (G2) 1 × 1; Technical analyst (G3) 2 × 1; Administrative support (G4) 1 × 1	Kenya data protection office

Each team was designed the way a coordinator would build it, with leadership, the specialists for each area and support staff. The supervision unit sits at the lower end of the range the European Commission estimated for the AI Act (1 to 25 full-time staff per country), and the sandbox and risk assessments follow the pace of Norway's sandbox (four projects a year, three to six months each).

Treatment of uncertainty

The model includes adjustments for the main sources of uncertainty in the estimate. The gap between the occupational group's average earnings and those of specialists is handled by the specialist premium; the age of the earnings data, by the inflation update; and the lack of national data, by ILOSTAT regional and world medians, always with the origin shown in each line's calculation.

Team composition, time shares and months were defined for each type of work and compared with public cases; these values may vary slightly with the context and can be adjusted role by role. When the same person coordinates more than one option, users only need to adjust the number of people or the coordinator's time share in each team. Cloud prices are those of a major public provider, used as a market reference; a national provider or a volume contract may have different prices, and the contract amount can replace the estimate. The sensitivity to the specialist premium in the Brazil test case is shown in the table below.

Specialist premium	Known cost of the test case (R$)	Change relative to the default
0%	5,494,260.12	−11.4%
25% (PwC, 2024 edition)	6,122,346.90	−1.2%
28% (default; Lightcast, 2025)	6,197,716.18	—
56% (PwC, 2025)	6,901,169.40	+11.4%

Confidence grades

Parameter	Grade	Source
Earnings by occupational group, country with data	A	ILOSTAT, by country and year
Consumer price inflation	A	IMF, World Economic Outlook
Employer contributions	A	SSA and ISSA, table 4
Exchange rate	A	ExchangeRate-API or entered in the case
IMF group, CGSI, population and education spending	A	IMF; World Bank (GTMI 2025, SE.XPD.TOTL.GB.ZS)
Earnings for a country without data	B	ILOSTAT regional, income-group or world median
Specialist premium	B	Lightcast (2025); compared with PwC (2025)
Travel, overhead and fee percentages	B	Global Fund, 2 CFR 200.414, UNSDG
Cloud prices and X-Road costs	B	Microsoft Azure (public price list); NIIS (X-Road calculators)
Facility operation and sovereign cloud premium	B	KPMG (2026); BCG (2025)
Team composition, time shares and months	C	Tool estimate, compared with public cases
Classes, days, instructors, travel and coordination of training	C	Tool estimate; development hours (ATD) and daily rates (UKCAMS) are grade B
Sovereign cloud coefficients (floor, part per 100,000 inhabitants, team, useful life)	C	Preliminary calibration with eight government cases
Suggested quantities for full-time staff (2 per profile), training (people in the plan, 20 hours, half in person) and the size of the challenge 2 systems	C	Tool estimate, so that no line is left without an amount

Worked example: Brazil test case

In the test case (Brazil, Brazilian reais, 12 months, 13 cost lines), the known cost is R$6,197,716.18: R$4,114,789.04 in options and R$2,082,927.14 in additional costs; category 3.1 is selected without an option and shows as pending, and training of people in the additional costs was not included. Brazil has its own ILOSTAT data (Continuous PNAD, 2025), updated to 2026 with the projected inflation of 4.4% (factor 1.044), and an employer contribution of 21%.

Group	Average monthly earnings, w (R$, 2025)	Monthly cost, c (R$)	Specialist monthly cost (R$)
G1 Managers	8,000.59	10,106.67	12,936.53
G2 Professionals	6,551.79	8,276.48	10,593.90

Group	Average monthly earnings, w (R$, 2025)	Monthly cost, c (R$)	Specialist monthly cost (R$)
G3 Technicians and associate professionals	4,332.05	5,472.42	7,004.70
G4 Clerical support workers	2,762.83	3,490.12	4,467.35

The option "AI safety and risk classification" has 8 months and six roles, two of them with the specialist premium. The team's monthly cost is R$48,190.75, and the option costs R$385,526.04.

$$\text{monthly cost} = 0.5 \times 10{,}106.67 + 2 \times 10{,}593.90 + 10{,}593.90 + 0.5 \times 8{,}276.48 + 5{,}472.42 + 0.5 \times 3{,}490.12 = 48{,}190.75$$

$$C_o = 8 \times 48{,}190.75 = 385{,}526.04$$

In challenge 2, the data inventory adds the 6-month team (R$183,174.99), two curators for the whole plan (2 × R$10,593.90 × 12 = R$254,253.60) and one virtual machine for the catalogue (US$0.115 × 730 × 12 × 5.2223 = R$5,260.95). Hosting 50 TB costs 50 × 1,024 × US$0.0208 × 12 × 5.2223 = R$66,738.49, and hosting and the catalogue virtual machine are usage. Local customisation, at R$353,026.20, is investment. Training (3.2), without entered quantities, trains the 62 people in the plan's teams for 20 hours, half in person: 3 classes, with 2 in-person days each. The specialist's hourly cost is R$10,593.90 × 12 ÷ 2,080 = R$61.12. Course creation costs (10 × 38 + 10 × 28) × R$61.12 = R$40,339.20, which is investment; instructors, 3 × 2 × 20 × 2 × R$61.12 = R$14,668.80; room and meals, (62 + 6) × 2 × R$188.78 = R$25,674.08, with the day delegate rate of £51 × 0.468 ÷ 0.873 converted at the pound exchange rate; instructors' travel, 3 × 2 × (2 × R$692.20 + R$244.40) = R$9,772.80; and coordination, 15% × R$50,115.68 = R$7,517.35. The total is R$97,972.23, and participants' time, outside the total, would be R$59,210.00.

Cat.	Option	Calculation	CapEx (R$)	OpEx (R$)	Usage (R$)	Cost (R$)
1.1	AI safety and risk classification	team, 8 months	0.00	385,526.04	0.00	385,526.04
1.1	Algorithmic fairness, non-discrimination, transparency, explainability and accountability	team, 9 months	0.00	508,205.11	0.00	508,205.11
1.2	Controlled testing environments	team, 12 months	0.00	613,225.68	0.00	613,225.68
1.2	Regulatory supervision	ongoing team	0.00	709,528.68	0.00	709,528.68
1.3	AI procurement and vendor evaluation guidelines	team, 5 months	0.00	145,720.14	0.00	145,720.14
1.3	Teacher and student AI-use policies	team, 4 months	0.00	82,476.99	0.00	82,476.99
2.1	Data inventory and curation	team, 6 months + curation + catalogue	0.00	437,428.59	5,260.95	442,689.54
2.2	Secure hosting and data storage	50 TB for 12 months	0.00	0.00	66,738.49	66,738.49
2.3	Open-source adoption and maintenance	ongoing team	0.00	397,541.82	0.00	397,541.82
2.3	Local customisation	team, 8 months	353,026.20	0.00	0.00	353,026.20
3.2	Training and professional development	62 people, 20 h, half in person, 3 classes	40,339.20	57,633.03	0.00	97,972.23
3.3	Governance committees	ongoing team	0.00	70,599.60	0.00	70,599.60
3.3	Interministerial coordination	ongoing team	0.00	241,538.52	0.00	241,538.52
	Direct cost, D		393,365.40	3,649,424.20	71,999.44	4,114,789.04
	People's work, P (D without hosting, catalogue and training)		353,026.20	3,591,791.17	0.00	3,944,817.37
	Travel, 25% × P		88,256.55	897,947.79	0.00	986,204.34
	Overhead, 15% × (P + V)		66,192.41	673,460.85	0.00	739,653.26

Cat.	Option	Calculation	CapEx (R$)	OpEx (R$)	Usage (R$)	Cost (R$)
	Administrative fee, 7% × (D + V)		34,135.12	316,686.52	6,247.90	357,069.54
	Known cost, C		581,949.48	5,537,519.36	78,247.34	6,197,716.18

With the three defaults, the additional costs add 50.6% to the direct cost. If the plan has no in-person activities, the institution already covers indirect costs and the funds are executed by a foundation charging 7%, the additional costs fall to R$288,035.23 (7% × D) and the known cost to R$4,402,824.27.

References

- ILO, ILOSTAT. Average monthly earnings of employees by sex, occupation and currency (DF_EAR_EMTA_SEX_OCU_CUR_NB). https://ilostat.ilo.org/data/
- IMF. World Economic Outlook: annual average consumer price inflation (PCPIPCH). Retrieved 9 November 2026. https://www.imf.org/external/datamapper/PCPIPCH@WEO
- SSA and ISSA. Social Security Programs Throughout the World, 2018–2019, table 4. https://www.ssa.gov/policy/docs/progdesc/ssptw/
- Lightcast. Beyond the Buzz: Developing the AI Skills Employers Actually Need, 2025. https://lightcast.io/resources/research/beyond-the-buzz-developing-the-ai-skills-employers-actually-need
- PwC. The Fearless Future: 2025 Global AI Jobs Barometer, 2025. https://www.pwc.com/gx/en/news-room/press-releases/2025/ai-linked-to-a-fourfold-increase-in-productivity-growth.html
- World Bank. Country classification by region and income group. https://api.worldbank.org/v2/country?format=json&per_page=400
- ExchangeRate-API. Daily exchange rates with USD base (the example uses those of 3 October 2026). https://open.er-api.com/v6/latest/USD
- Fawaz Ahmed. currency-api: daily exchange rates with USD base. https://github.com/fawazahmed0/exchange-api
- Global Fund. Data Explorer: budgets by cost category, cycles 5 to 7. https://data.theglobalfund.org/financial-insights
- United States. 2 CFR 200.414, indirect costs, 2024 revision. https://www.ecfr.gov/current/title-2/subtitle-A/chapter-II/part-200/subpart-E/subject-group-ECFRd93f2a98b1f6455/section-200.414
- UN Energy MPTF. 2025 annual financial report (7% indirect costs, UNSDG policy). https://mptf.undp.org/sites/default/files/documents/2026-05/2025_annual_financial_report-un_energy_mptf.pdf
- European Commission. AI Act proposal, COM(2021) 206, section 4. https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:52021PC0206
- Cambridge Forum on AI: Law and Governance. From watchdogs to partners in tech innovation: Norway's AI sandbox. https://www.cambridge.org/core/journals/cambridge-forum-on-ai-law-and-governance/article/from-watchdogs-to-partners-in-tech-innovation-navigating-regulatory-roles-in-norways-ai-sandbox/7E2FA0B1F75E6803B2DE4124EB0FC6CB
- Open Budget Kenya. Office of the Data Protection Commissioner, 2021–2025. https://openbudget.or.ke/project/1122002100_The_Office_of_the_Data_Protection_Commi/2024
- Microsoft Azure. Retail Prices API: public price list, retrieved 10 November 2026. https://prices.azure.com/api/retail/prices
- Google Cloud. Cloud Storage pricing. https://cloud.google.com/storage/pricing
- NIIS. X-Road cost calculators, 2021. https://x-road.global/xroad-cost-calculators
- CISA. Back Up Government Data (3-2-1 rule). https://www.cisa.gov/audiences/state-local-tribal-and-territorial-government/secure-us-sltt/back-government-data
- European Union. Regulation (EU) 2023/2854 (Data Act), art. 29. https://eur-lex.europa.eu/eli/reg/2023/2854/oj
- World Bank. GovTech Maturity Index 2025. https://www.worldbank.org/en/programs/govtech/gtmi-2025-update
- World Bank. Government expenditure on education, % of government expenditure (SE.XPD.TOTL.GB.ZS). https://data.worldbank.org/indicator/SE.XPD.TOTL.GB.ZS
- IMF. World Economic Outlook, April 2025: country groups. https://meetings.imf.org/en/-/media/files/publications/weo/2025/april/english/statsappendix.pdf

- KPMG. Benchmarking CapEx and OpEx in the Global Data Centre Market, 2026. https://kpmg.com/ie/en/insights/strategy/global-data-centre-cost-benchmarks.html
- BCG. Cloud Cover: Price, Sovereignty Demands, and Waste, 2025. https://www.bcg.com/publications/2025/cloud-cover-price-sovereignty-demands-waste
- ODC-Noord. Inzending ODC-Noord, 2026. https://www.futuregov.nl/wp-content/uploads/2026/09/Inzending-ODC-Noord-1.0-2.pdf
- ATD. How Long to Develop One Hour of Training? Updated for 2017. https://www.td.org/content/atd-blog/how-long-does-it-take-to-develop-one-hour-of-training-2017
- UK Events. UK Conference and Meeting Survey (UKCAMS) 2026. https://ukevents.org.uk/research-and-publications/research/industry-research/ukcams-2026
- World Bank. PPP conversion factor, private consumption (PA.NUS.PRVT.PP), and official exchange rate (PA.NUS.FCRF). Retrieved 10 November 2026. https://data.worldbank.org/indicator/PA.NUS.PRVT.PP
- World Bank. Enrolment in primary (SE.PRM.ENRL) and secondary (SE.SEC.ENRL) education. Retrieved 10 November 2026. https://data.worldbank.org/indicator/SE.PRM.ENRL
- IAPP. AI Governance Professional (AIGP) Online Training (market reference). https://store.iapp.org/ai-governance-professional-aigp-online-training/